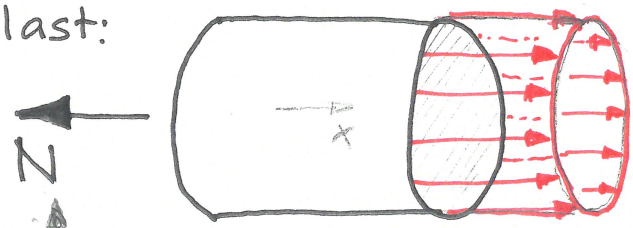
Olika sorters laster - Spänningar på snittytor

Dragen stång:

Stångkraft / dragspänning

Snitt-last:



resultant

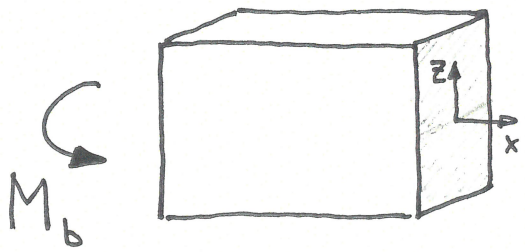
$$\sigma = \frac{N}{A}$$

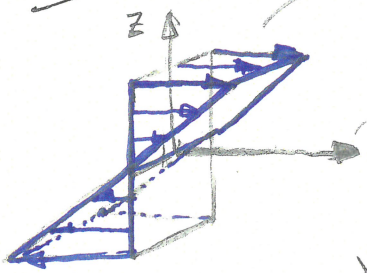
Normalkraft (drag)

tvärsnittsarea

Böjd balk:

Böjmoment / böjspänningar



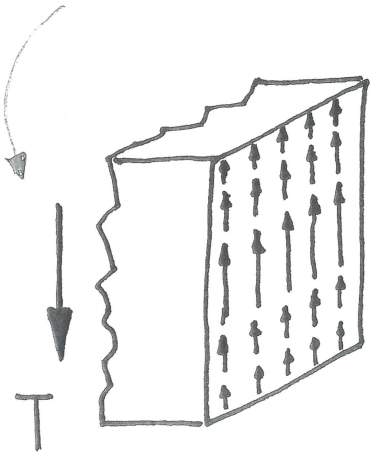


med tecken

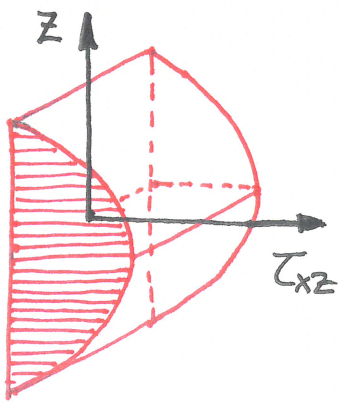
$$\sigma_b(z) = \frac{M_b}{I} \cdot z$$

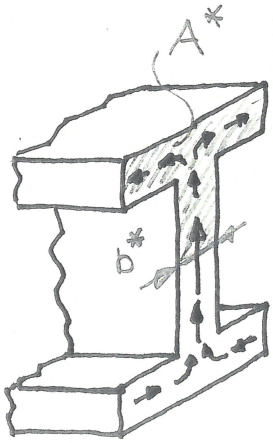
I - tröghetsmoment (Area-)

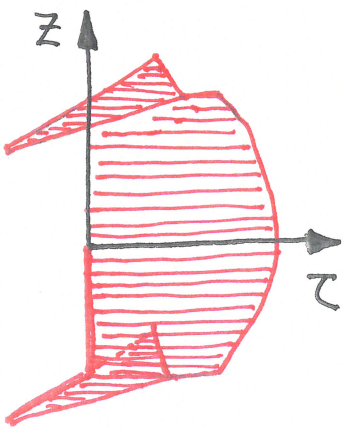
Tvärkraft i balk - Böjskjuvning : $\tau = \frac{S(A^*) \cdot T}{I b^*}$



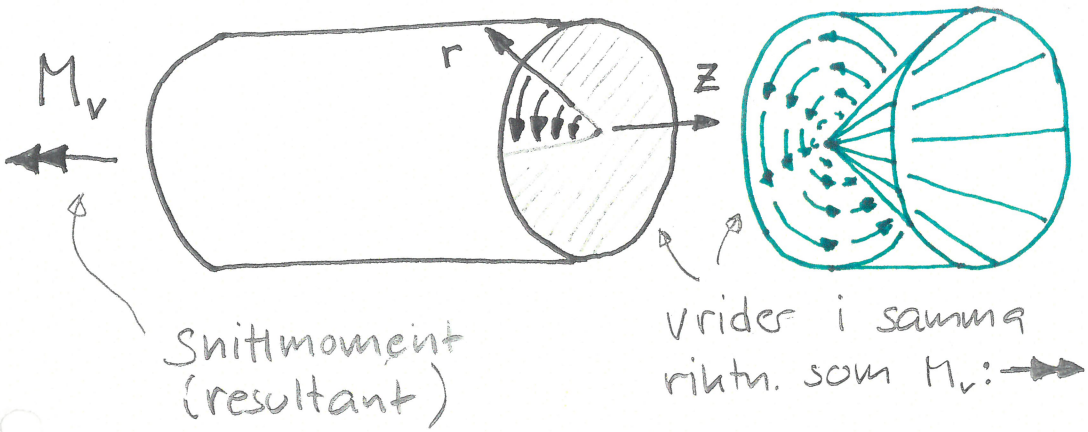
(resultant)







Vriden axel: Vridmoment - vridskjuvspänning



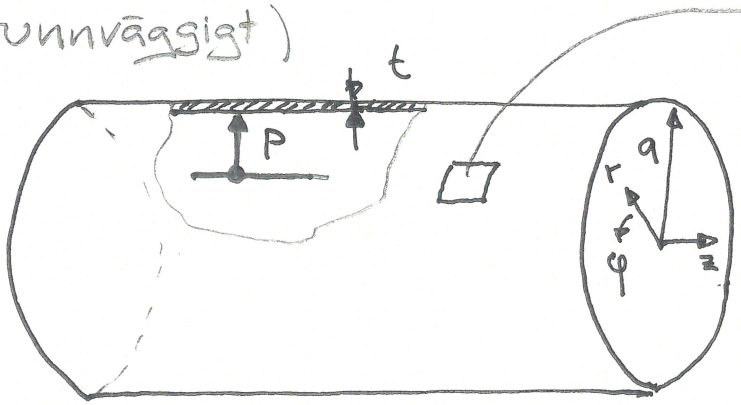
$$\tau_v(r) = \frac{M_v}{K} \cdot r$$

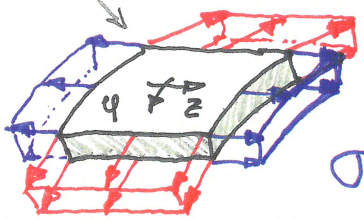
K: Vridstyvhetens tvärsnittsfaktor

Olika laster - Tryckkärl / fleraxliga

Cylindriskt:

(tunnväggigt)



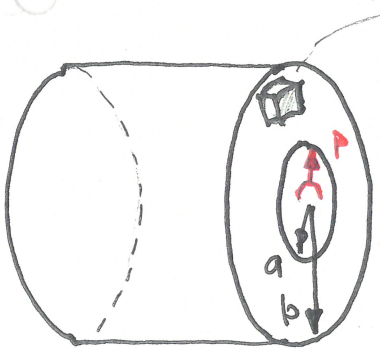


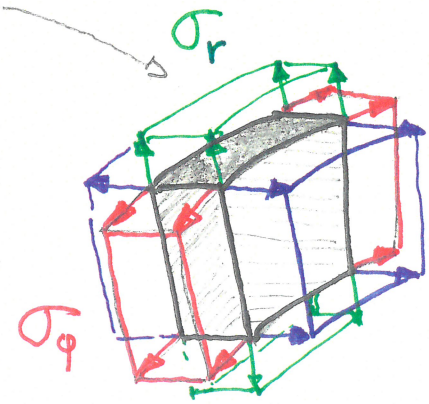
$$\sigma_\varphi = \frac{pa}{t}$$

$$\sigma_z = \frac{pa}{2t}$$

$(\sigma_r \approx 0)$

Tjockväggigt rör / skiva (cirkulärsymm.)





$$\left.\begin{aligned} \sigma_r &= A - \frac{B}{r^2} \\ \sigma_\varphi &= A + \frac{B}{r^2} \end{aligned}\right\} \text{Ges RV}$$

A & B

σ_z - konstant (best. av jvkt i z-led)

polära koordinater

